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Deep Learning vs. Machine Learning – What's The Difference?

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To most people, the terms Deep Learning and Machine Learning seem like interchangeable buzzwords in the AI world. However, that's not true. Hence, everyone who seeks to better understand the field of

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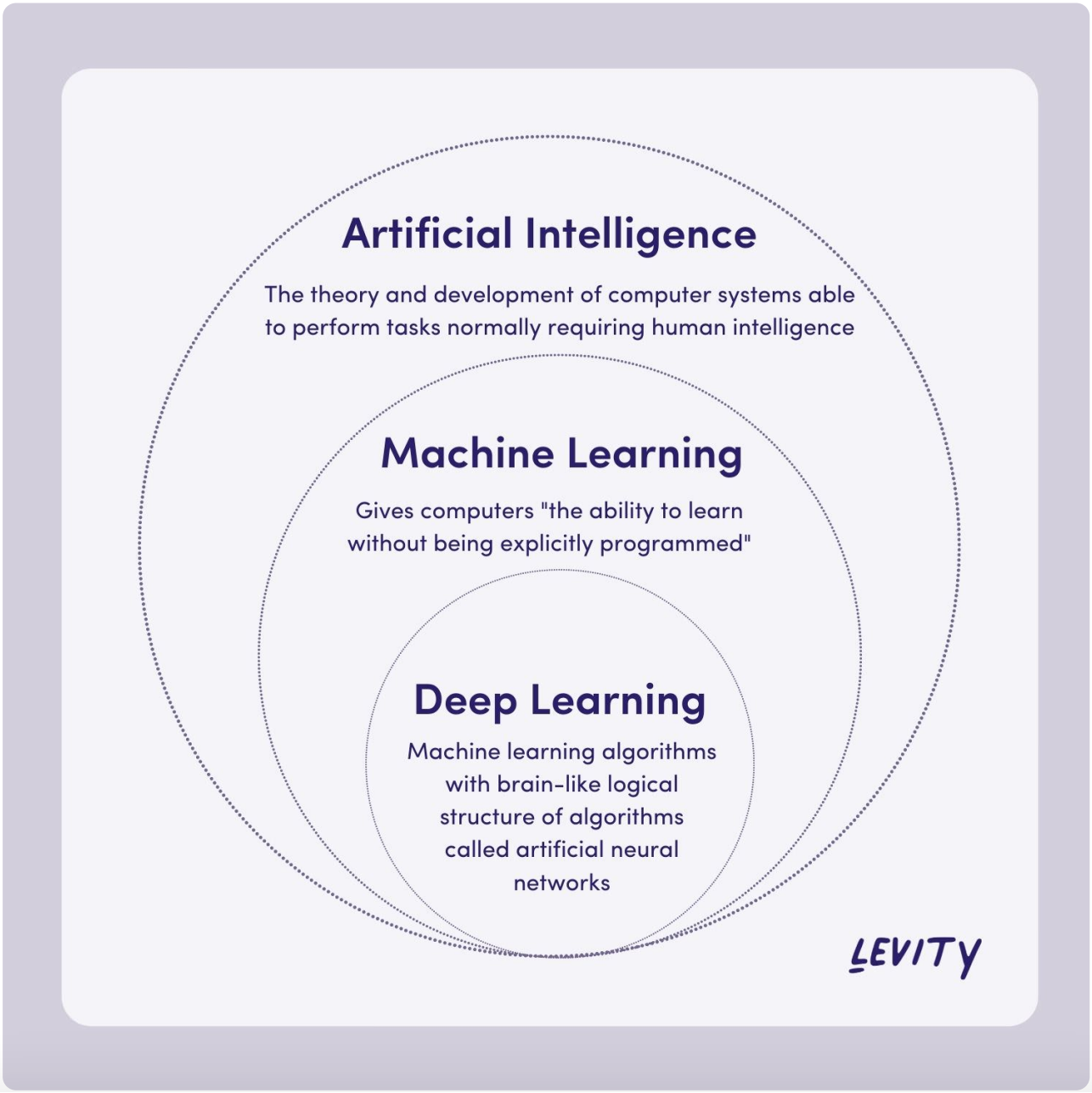
What's the difference between Deep Learning and Machine Learning?

Machine Learning means computers learning from data using algorithms to perform a task without being explicitly programmed. Deep Learning uses a complex structure of algorithms modeled on the human brain. This enables the processing of unstructured data such as documents, images, and text.



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To break it down in a single sentence: Deep Learning is a specialized subset of Machine Learning which, in turn, is a subset of Artificial Intelligence. In other words, Deep Learning *is* Machine Learning.

But let's dig a little bit deeper.

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Machine Learning vs. Deep Learning [What's the difference?]

What is Machine Learning?

Machine Learning is the general term for when computers learn from data. It describes the intersection of computer science and statistics where algorithms are used to perform a specific task without being

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In general, the learning process of these algorithms can either be

read through this article.

A traditional Machine Learning algorithm can be something as simple as linear regression. For instance, imagine you want to predict your income given your years of higher education. In the first step, you have to define a function, e.g. income = y + x * years of education. Then, give your algorithm a set of training data. This could be a simple table with data on some people's years of higher education and their associated income. Next, let your algorithm draw the line, e.g. through an ordinary least squares (OLS) regression. Now, you can give the algorithm some test data, e.g. your personal years of higher education, and let it predict your income.

While this example sounds simple it does count as Machine Learning – and yes, the driving force behind Machine Learning is ordinary statistics. The algorithm learned to make a prediction without being explicitly programmed, only based on patterns and inference.

So much about Machine Learning in general – to summarize:

- Machine Learning is at the intersection of computer science and statistics through which computers receive the ability to learn without being explicitly programmed.
- There are two broad categories of Machine Learning problems: supervised and unsupervised learning.
- A Machine Learning algorithm can be something as simple as an OLS regression.

Let's now examine how the term Deep Learning relates to all of this.

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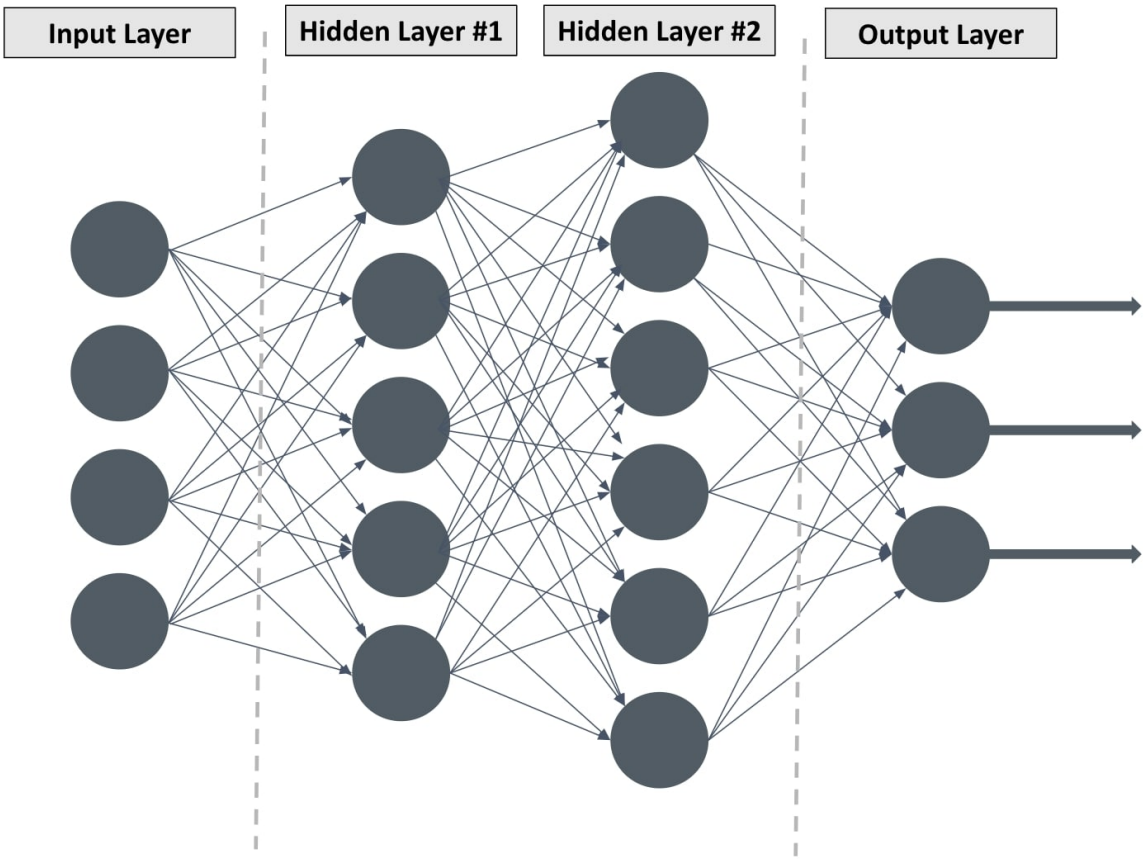
Deep Learning algorithms can be regarded both as a sophisticated and mathematically complex evolution of machine learning algorithms. The field has been getting lots of attention lately and for good reason: Recent developments have led to results that were not thought to be possible before.

Deep Learning describes algorithms that analyze data with a logical structure similar to how a human would draw conclusions. Note that this can happen both through supervised and unsupervised learning. To achieve this, Deep Learning applications use a layered structure of algorithms called an artificial neural network (ANN). The design of such an ANN is inspired by the biological neural network of the human brain, leading to a process of learning that's far more capable than that of standard machine learning models.



A simple artificial neural network

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middle layers are called hidden layers because their values aren't observable in the training set. In simple terms, hidden layers are calculated values used by the network to do its "magic". The more hidden layers a network has between the input and output layer, the *deeper* it is. In general, any ANN with two or more hidden layers is referred to as a deep neural network.

Today, Deep Learning is used in many fields. In automated driving, for instance, Deep Learning is used to detect objects, such as STOP signs or pedestrians. The military uses Deep Learning to identify objects from satellites, e.g. to discover safe or unsafe zones for its troops. Of course, the consumer electronics industry is full of Deep Learning, too. Home assistance devices such as Amazon Alexa, for example, rely on Deep Learning algorithms to respond to your voice and know your preferences.

How about a more concrete example? Imagine the company Tesla using a Deep Learning algorithm for its cars to recognize STOP signs. In the first step, the ANN would identify the relevant properties of the STOP sign, also called *features*. Features may be specific structures in the inputted image, such as points, edges, or objects. While a software engineer would have to select the relevant features in a more traditional Machine Learning algorithm, the ANN is capable of *automatic feature engineering*. The first hidden layer might learn how to detect edges, the next is how to differentiate colors, and the last learn how to detect more complex shapes catered specifically to the shape of the object we are trying to recognize. When fed with training data, the Deep Learning algorithms would eventually learn from their own errors whether the prediction was good, or whether it needs to adjust.

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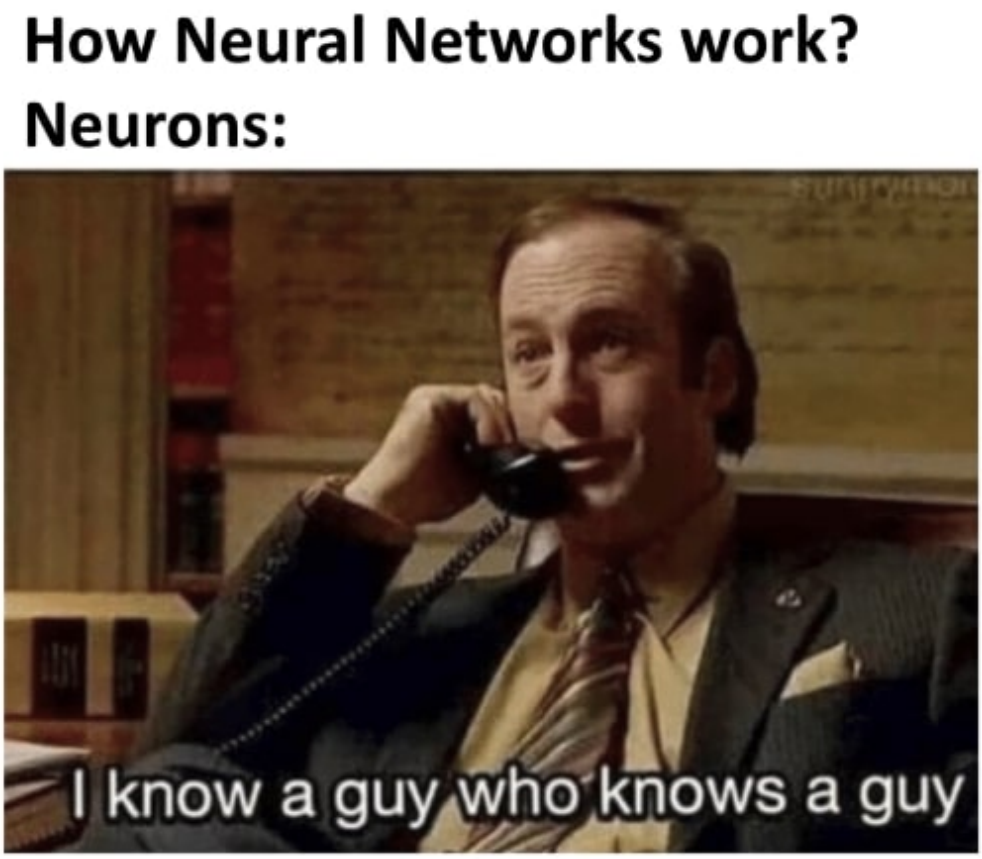

What the job of a neuron in an ANN really looks like.

Overall, through automatic feature engineering and its self-learning capabilities, the Deep Learning algorithms need only little human intervention. While this shows the huge potential of Deep Learning, there are two main reasons why it has only recently attained so much usability: data availability and computing power.

Firstly, Deep Learning requires incredibly vast amounts of data (we will get to exceptions to that rule). Tesla's autonomous driving software, for instance, needs millions of images and video hours to function properly.

Secondly, Deep Learning needs substantial computing power. However, with the emergence of cloud computing infrastructure and high-performance GPUs (graphic processing units, used for faster calculations) the time for training a Deep Learning network could be

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trained models. The reason: Transfer learning can be regarded as a cure for the needs of large training datasets that were necessary for ANNs to produce meaningful results.

These enormous data needs used to be the reason why ANN algorithms weren't considered to be the optimal solution to all problems in the past. However, for many applications, this need for data can now be satisfied by using pre-trained models. In case you want to dig deeper, we recently published an article on transfer learning.

To sum up:

- Deep Learning is a specialized subset of Machine Learning.
- Deep Learning relies on a layered structure of algorithms called an artificial neural network.
- Deep Learning has huge data needs but requires little human intervention to function properly.
- Transfer learning is a cure for the needs of large training datasets.

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The main differences between Machine Learning and Deep Learning

This is a common question and if you have read this far, you probably know by now that it should not be asked in that way. Deep Learning algorithms *are* Machine Learning algorithms. Therefore, it might be better to think about what makes Deep Learning *special* within the field of Machine Learning. The answer: the ANN algorithm structure, the lower need for human intervention, and the larger data requirements.

First and foremost, while traditional Machine Learning algorithms have a rather simple structure, such as linear regression or a decision tree, Deep Learning is based on an artificial neural network. This multi-layered ANN is, like a human brain, complex and intertwined.

Secondly, Deep Learning algorithms require much less human intervention. Remember the Tesla example? If the STOP sign image recognition was a more traditional machine learning algorithm, a software engineer would manually choose features and a classifier to sort images, check whether the output is as required, and adjust the algorithm if this is not the case. As a deep learning algorithm, however, the features are extracted automatically, and the algorithm learns from its own errors (see image below).

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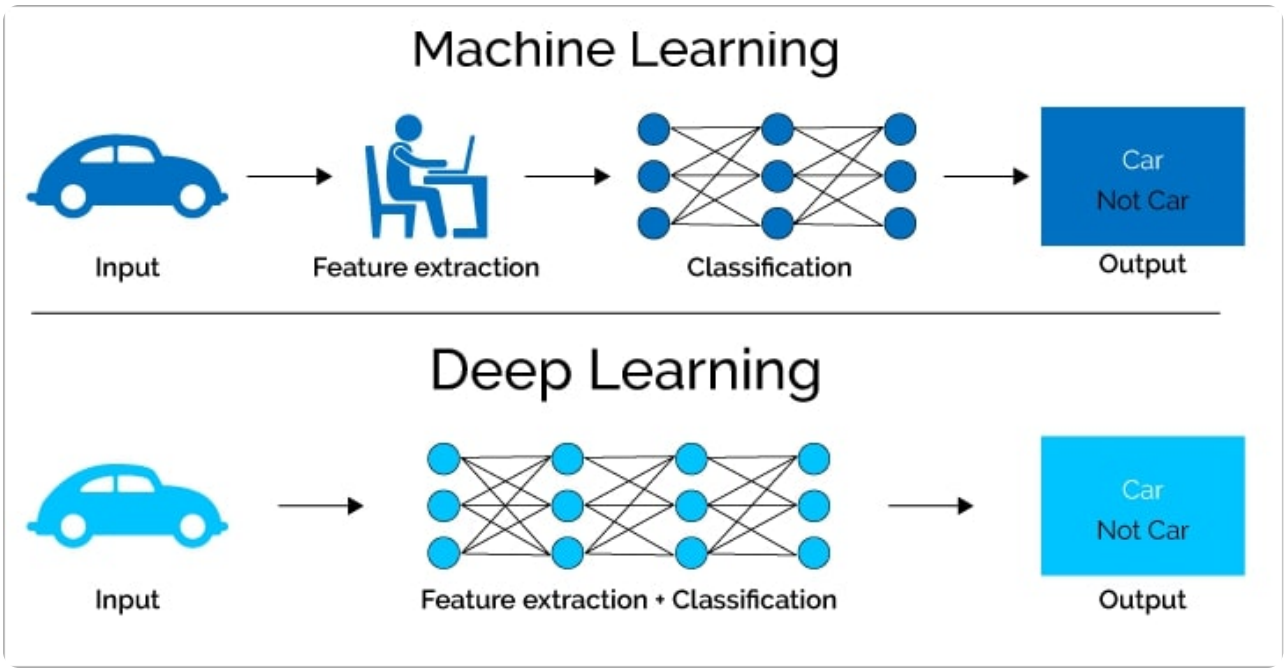

The Deep Learning algorithm doesn't need a software engineer to identify features but is capable of automatic feature engineering through its neural network. (Source: softwaretestinghelp.com)

Thirdly, Deep Learning requires much more data than a traditional Machine Learning algorithm to function properly. Machine Learning works with a thousand data points, deep learning oftentimes only with millions. Due to the complex multi-layer structure, a deep learning system needs a large dataset to eliminate fluctuations and make high-quality interpretations.

Find out more about Data Science vs ML vs AI

Got it. But what about coding?

Deep Learning is still in its infancy in some areas but its power is already enormous. It is mostly leveraged by large companies with vast financial and human resources since building Deep Learning algorithms used to be complex and expensive. But this is changing. We at Levity believe that everyone should be able to build his own custom

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